UNIT-4: Matrices & Successive Differentiation
Appearance Year: 2025-26

1. Rank & Systems of Linear Equations:
   - Find Rank of Matrix A = [[2,-4,6], [-1,2,-3], [3,-6,9]].
   - Solve matrix equation [[3,-1],[2,5]][x, y]^T = [[4],[-3]].
   - Determine lambda & mu for system solutions consistency.
2. Eigen Values & Vectors:
   - Find eigen values & vectors of A = [[6,-2,2], [-2,3,-1], [2,-1,3]].
3. Successive Differentiation & Curve Tracing:
   - If y = x^3 log(x), prove x(d^4y/dx^4) = 6/x.
   - Find nth derivative of (2x+1)/((2x-1)(2x+3)).
   - Trace curve y^2(a+x) = x^2(3a-x).